

Section Report 3 — Independent Amplicon Analysis

BIOL 4810

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Section Report 3 — Independent Amplicon Analysis

Goal: Use your **own sequencing dataset** to complete the full workflow in Galaxy: process amplicon reads with **DADA2**, then visualize your results in **phyloseq** using **alpha diversity**, **beta diversity**, and a **taxonomic bar plot**.

Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

0) Your dataset

0A. Sample set / project name:

0B. Sequencing type:

0C. Metadata grouping variable you analyzed:

Workflow overview

For this section report, you will complete:

```
plotQualityProfile → filterAndTrim → learnErrors → dada → mergePairs →  
makeSequenceTable → removeBimeraDenovo → assignTaxonomy → phyloseq alpha  
diversity → phyloseq beta diversity (NMDS using Bray-Curtis) → phyloseq  
taxonomic bar plot
```

Reminder: This section report asks you to apply the entire workflow to your own data and explain the biological meaning of each major output.

Part 1 — Quality profiles

1A) Forward read quality profile

What is the purpose of this step?

Upload screenshot: forward plotQualityProfile PDF plotQuality_F.png**Where does quality begin to noticeably decline in the forward reads?****Chosen forward truncation length:**

1B) Reverse read quality profile

What is usually different about reverse-read quality compared with forward-read quality?**Upload screenshot: reverse plotQualityProfile PDF** plotQuality_R.png**Where does quality begin to noticeably decline in the reverse reads?****Chosen reverse truncation length:**

1C) Truncation decision

Explain why you chose these truncation values.

Rule of thumb: trim where quality begins to drop, but still keep enough overlap for forward and reverse reads to merge.

Part 2 — filterAndTrim

What filterAndTrim does:

This step removes low-quality reads and trims poor-quality bases from the ends of reads before denoising.

Why is this step important?

This step is important because it helps to trim the low-quality reads since they would impact the results in the future.

Parameters used:

Truncation at 230 for the reverse reads only. The front of those reads were good, it was just the end where it got a little iffy.

Upload screenshot: filterAndTrim output

No file chosen

What could happen if you trim too aggressively?

If you trim too aggressively, some tools may not work properly since there's not enough overlap (maybe speaking from experience)

What could happen if you do not trim enough?

If you don't trim enough, the quality could be worse in the long run.

IMPORTANT — Split Paired Reads (Unzip Collection)

After `filterAndTrim`, your output is a **paired collection**.

Before continuing, you must split this into **forward and reverse collections**.

What to do in Galaxy

1. In the Tools panel, search: **unzip collection**
2. Input: your `filterAndTrim` paired output
3. Run

What you will get

- One collection = **forward reads (R1)**
- One collection = **reverse reads (R2)**

Why this matters

`learnErrors` and `dada` must be run **separately** on forward and reverse reads.

If you skip this, `mergePairs` will fail.

Part 3 — learnErrors

What learnErrors does:

This step models the sequencing error rates in your dataset so DADA2 can distinguish real sequence variants from sequencing mistakes.

Why is this step important?

LearnErrors makes it so DADA2 can recognize any sequencing errors, which can be seen on the plots it produces.

Upload screenshot: learnErrors PDF

learnErrors_F.png

Does the fitted model generally follow the observed error pattern?

Mostly yes

Why is learning error rates important before denoising?

So learning error rates is important before denoising because DADA2 relies on the error models to see if the differences are caused by actual biological variation or by sequencing errors.

Part 4 — dada (denoising)**What dada does:**

This is the denoising step. DADA2 uses the learned error model to correct likely sequencing errors and infer exact amplicon sequence variants (ASVs).

What is denoising doing?

Denoising is correcting sequencing errors in the reads to identify and remove any substitutions, insertions, and any other mistakes introduced during sequencing.

How is ASV inference different from OTU clustering?

ASV inference is different from OTU clustering because ASV identifies sequence variants at single-nucleotide places whereas OTU clusters groups based on similarity.

Part 5 — mergePairs**What mergePairs does:**

This step merges the forward and reverse reads for each amplicon so you can reconstruct the full sequence from overlapping read pairs.

What is being merged, and why?

The forward and reverse reads are being merged to improve accuracy as they will have overlapping regions. This will allow errors to be detected and corrected to get a more reliable final sequence.

Why do forward and reverse reads need to overlap?

The forward and reverse reads need to overlap to correct sequencing errors.

If reads fail to merge, what is one likely reason?

If reads fail to merge, one likely reason is that there is not enough overlap between the reads.

Part 6 — makeSequenceTable**What makeSequenceTable does:**

This step creates the ASV table. Each row is a sample, each column is an ASV, and each cell contains the number of times that ASV was observed in that sample.

What does this table represent biologically?

This table contains which organisms are present in the samples and their relative abundance.

What are the rows, columns, and cell values in the sequence table?

The rows are individual samples. The columns are ASVs. The cell values represent the number of times an ASV was observed in the sample.

Upload screenshot: sequence table output

sequenceTable.csv

Upload screenshot: sequence length distribution

sequenceTable_plot.png

What does the sequence length distribution show?

The sequence length distributions shows a clean and short amplicon. It does have a bit more bars down the length, which suggests that there may be minor sequencing variation. It could also suggest chimeras.

Part 7 — removeBimeraDenovo**What removeBimeraDenovo does:**

This step removes chimeras, which are artificial sequences created during PCR when fragments from different templates are incorrectly joined together.

What is a chimera, and why should it be removed?

A chimera is an artificial DNA sequence formed during PCR when fragments from different sequences are mistakenly joined together. This means that it should be removed because it doesn't actually represent a

Why are chimeras a problem for interpretation?

Chimeras are a problem for interpretation because they mess with the results when they get picked up. Since they aren't actually any real biological organisms, they just create problems.

How many chimeric reads were removed? How do you know?

Around 36 chimeric reads were removed. I know because I checked the sequence table and it had 2020 lines, while post-RemoveBimeraDenovo it had 1984 lines.

Part 8 — assignTaxonomy**What assignTaxonomy does:**

This step compares your ASVs to a reference database and assigns taxonomy, such as phylum, class, order, family, genus, or sometimes species.

What does taxonomy assignment add to your ASV table?

Taxonomy assignment adds biological meaning to the ASV table because it links each DNA sequence to known groups of organisms, like genus, species, phylum, etc.

Why might some ASVs not be assigned all the way to species?

Some ASVs might not be assigned all the way to species because the reference database may not have an exact match.

Part 9 — Alpha diversity**9A) Alpha diversity****In your own words, what is alpha diversity?**

Alpha diversity is a metric to represent to diverse a community is.

9B) Observed richness**What does Observed measure?**

Observed measures the total number of species.

9C) Shannon diversity

What does Shannon measure?

Shannon measures species abundance between sets.

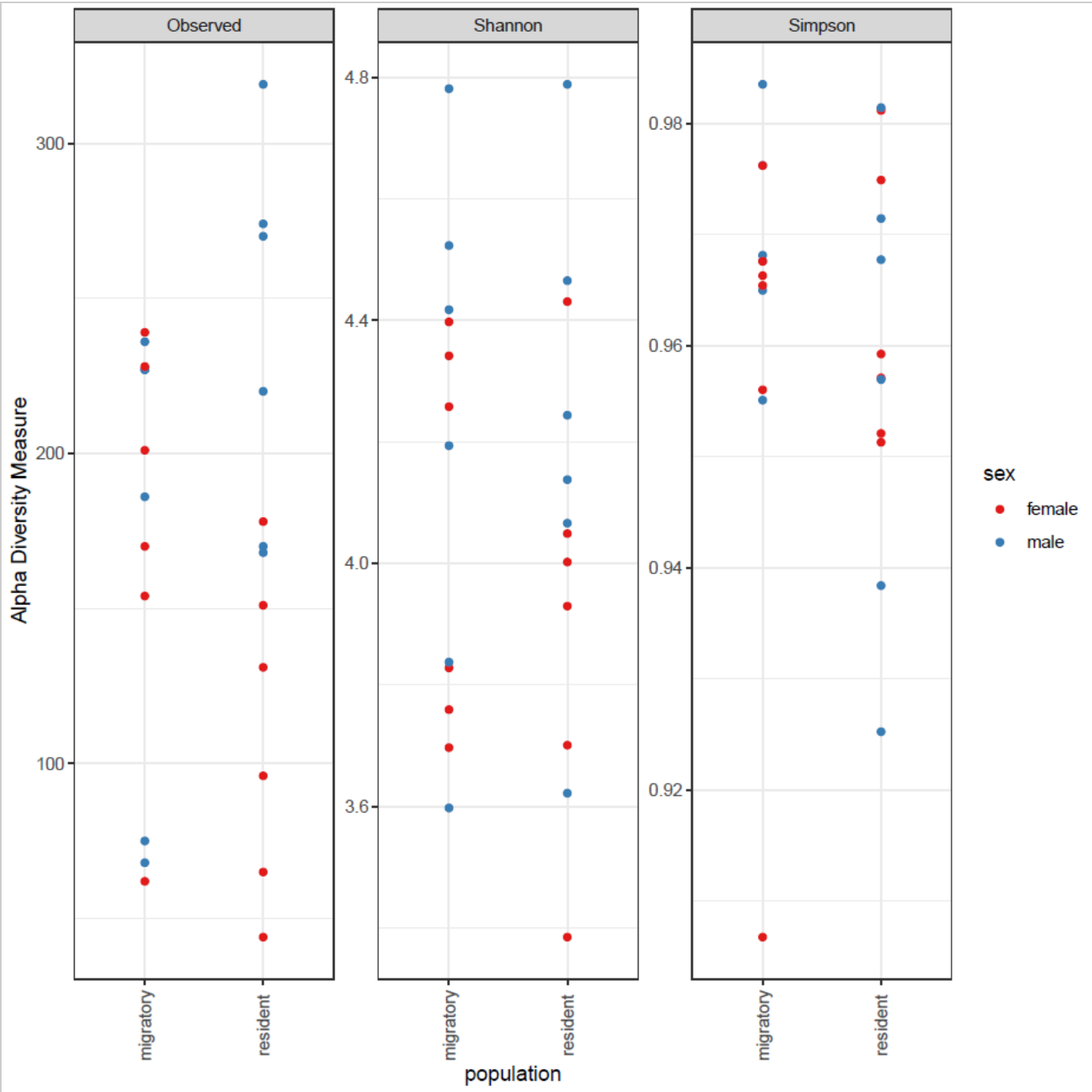
9D) Simpson diversity

What does Simpson measure?

Simpson measures works like Shannon, but is less sensitive to rare species.

Upload screenshot: alpha diversity plot grouped by population

Choose File alphaDiversi...pulation.png

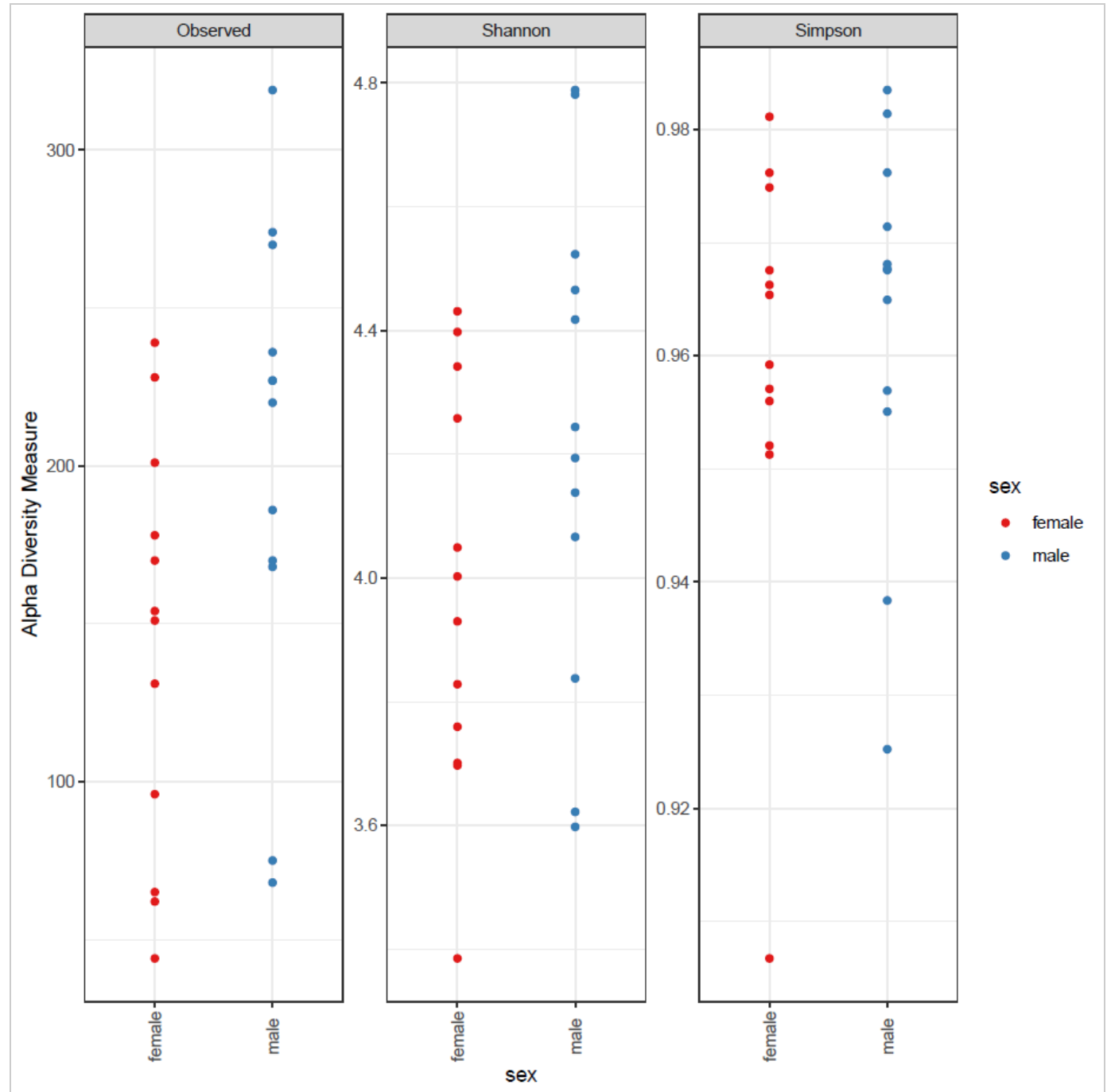


Describe differences in alpha diversity by population.

Across all three metrics, migratory and resident populations show a fairly similar density overall. There is a lot of overlap between the two groups, but migratory samples have a sometimes higher Shannon and Simpson

Upload screenshot: alpha diversity plot grouped by sex

Choose File alphaDiversity_sex.png

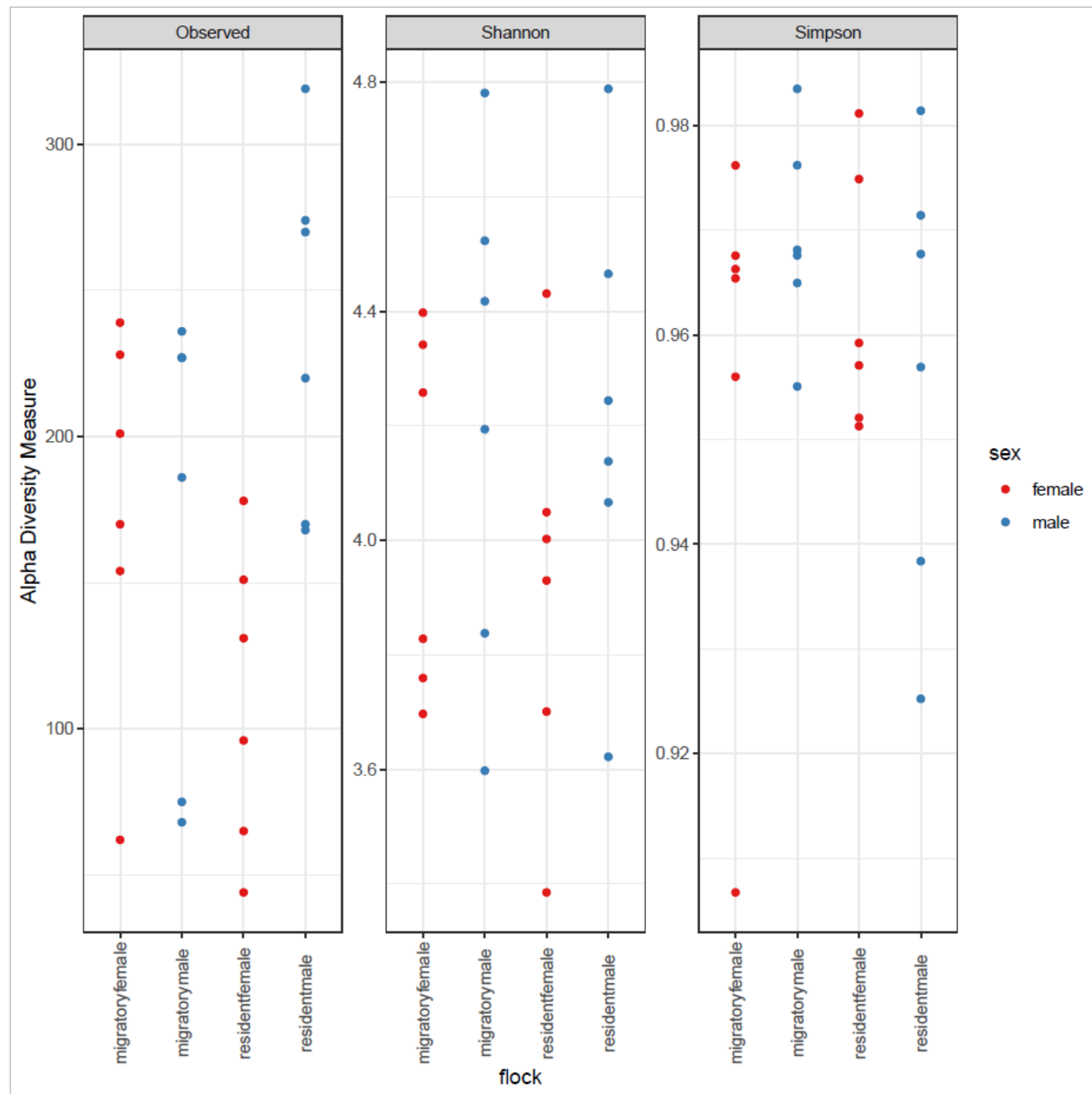


Describe differences in alpha diversity by sex.

Comparing alpha diversity by sex, there does seem to be a more noticeable difference. Males tend to have higher Observed richness, and slightly higher Shannon diversity compared to females. Simpson values are

Upload screenshot: alpha diversity plot grouped by flock

Choose File alphaDiversity_plot.png



Describe differences in alpha diversity by flock.

Looking at the flock, resident males show the highest diversity in Observed richness. Migratory males are also relatively high, but they seem to be slightly lower than resident males. Females (both migratory and resident)

Part 10 — Beta diversity (NMDS with Bray-Curtis)

What beta diversity means:

Beta diversity describes differences in community composition **between samples**.

In your own words, what is beta diversity?

Beta diversity is comparing the composition of one community to another.

10A) Bray-Curtis dissimilarity**What does Bray-Curtis measure?**

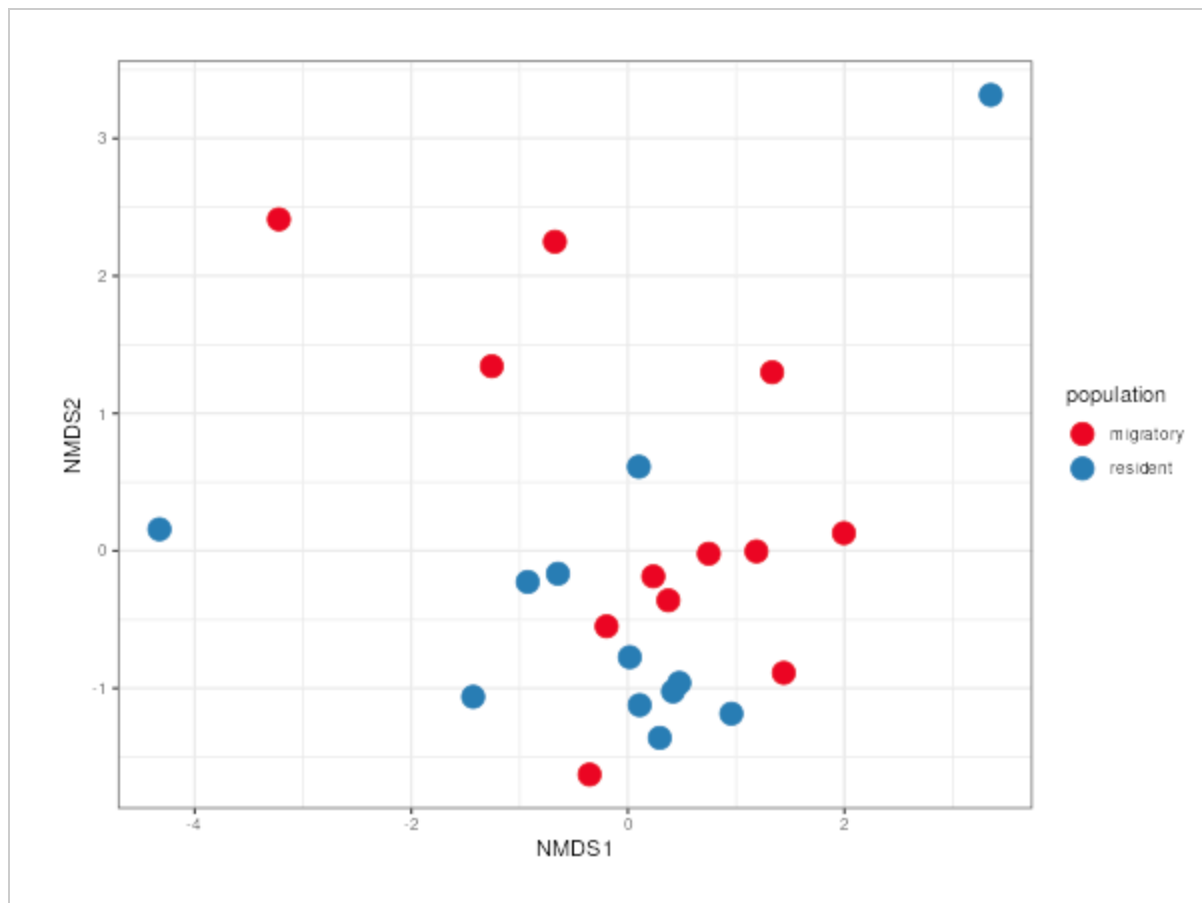
Bray-Curtis measures beta diversity, so if this metric is higher then there are no shared species. If it's lower, then there are more shared species.

10B) NMDS**What does NMDS show?**

NMDS shows each community present and its distance from every other point.

Upload screenshot: NMDS Bray-Curtis plot colored by population

Choose File NMDS_population.png

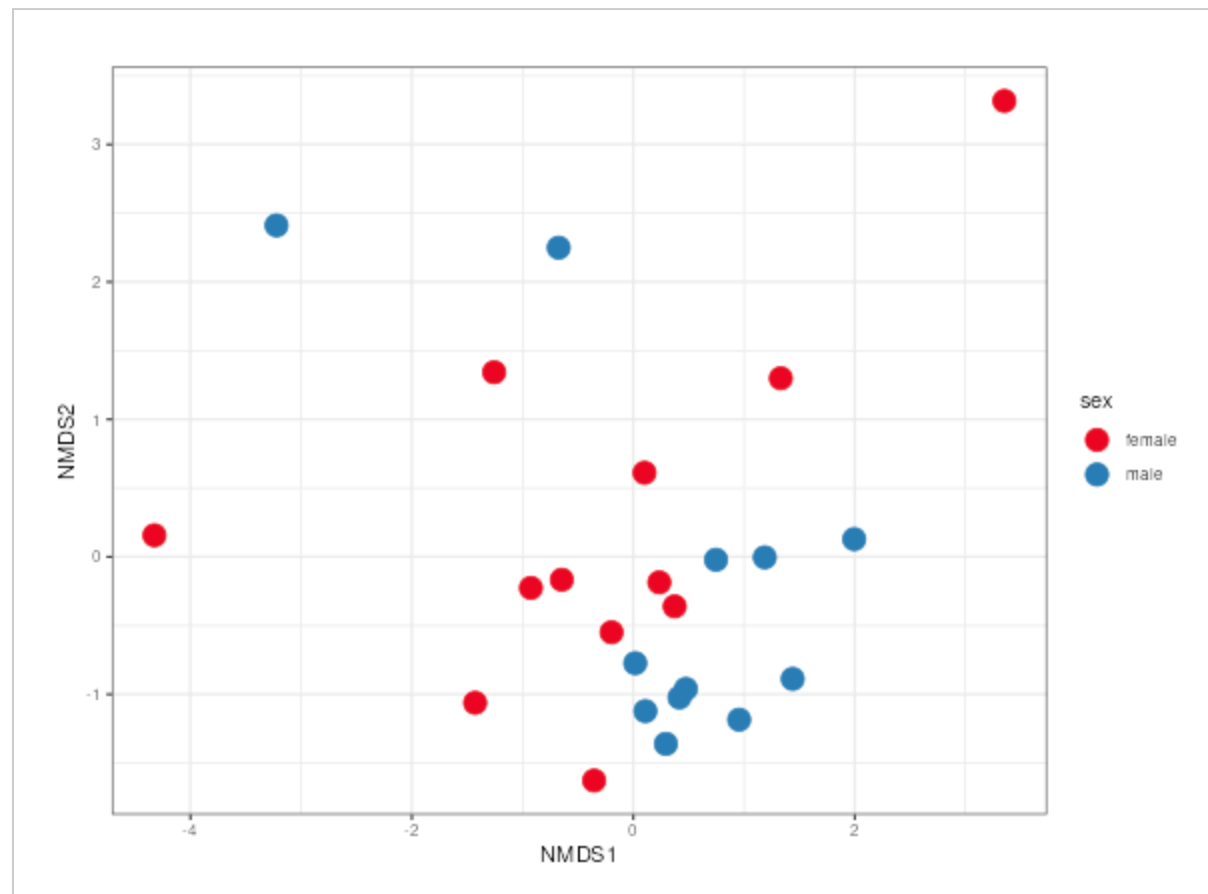


Describe beta diversity patterns by population.

Migratory and resident samples show substantial overlap in ordination space, showing that their microbial communities are not strongly separated. While there is some loose clustering, the groups seem to be largely

Upload screenshot: NMDS Bray-Curtis plot colored by sex

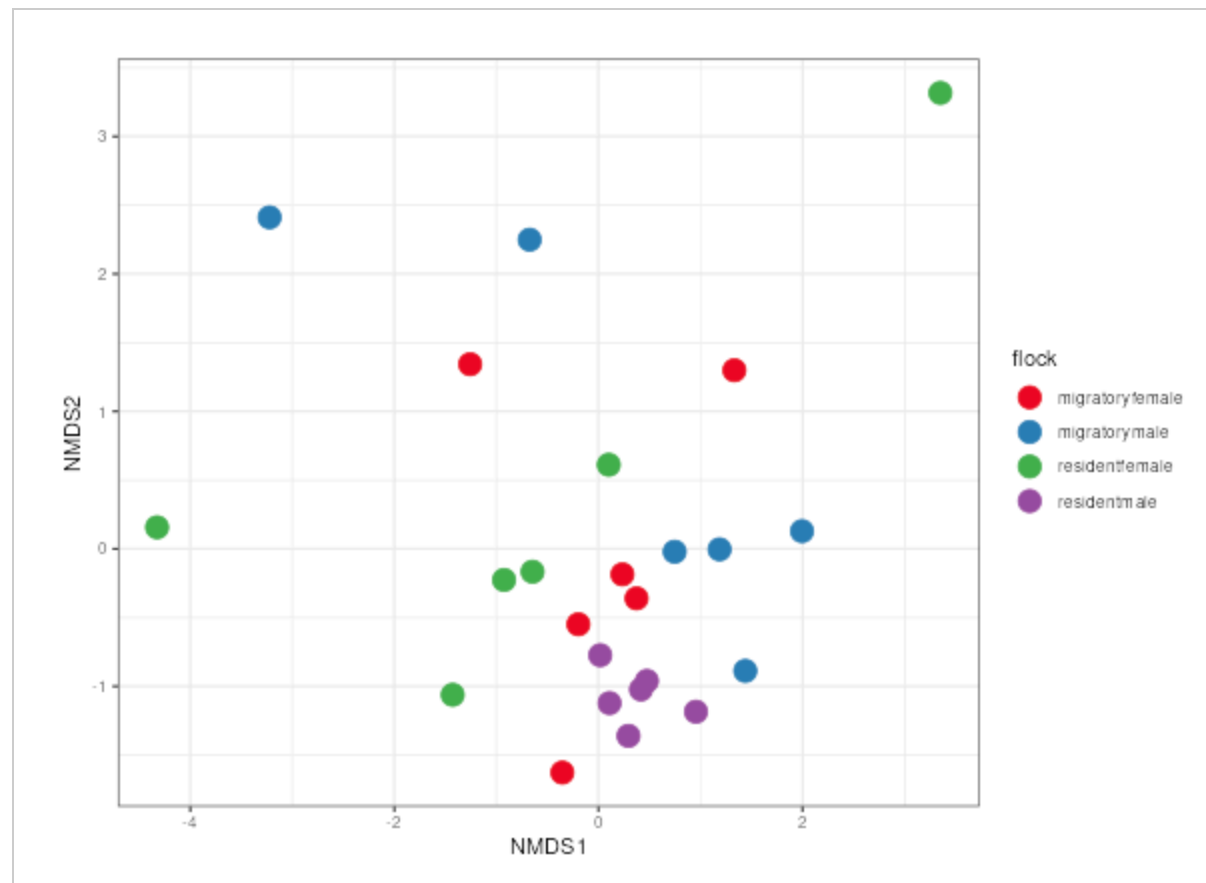
Choose File NMDS_sex.png

**Describe beta diversity patterns by sex.**

Compared to population, there is moderate separation by sex: male samples tend to cluster more tightly in one region (lower NMDS2 value, slightly higher NMDS1), while females are more dispersed and cover a wider range of the plot. This could potentially suggest that sex may influence microbial composition.

Upload screenshot: NMDS Bray-Curtis plot colored by flock

Choose File NMDS_flock.png

**Describe beta diversity patterns by flock.**

Looking at the flock, resident males seem to be the tightest cluster, indicating very similar community composition. The migratory males are somewhat spread, but four points show partial grouping. Females, both

Part 11 — Taxonomic bar plot**What a taxonomic bar plot shows:**

A bar plot summarizes the composition of samples by showing the relative abundance of taxa at a selected taxonomic level.

What taxonomic level did you use?

Phylum

How did you facet the plot?

sex

Why was this taxonomic level useful?

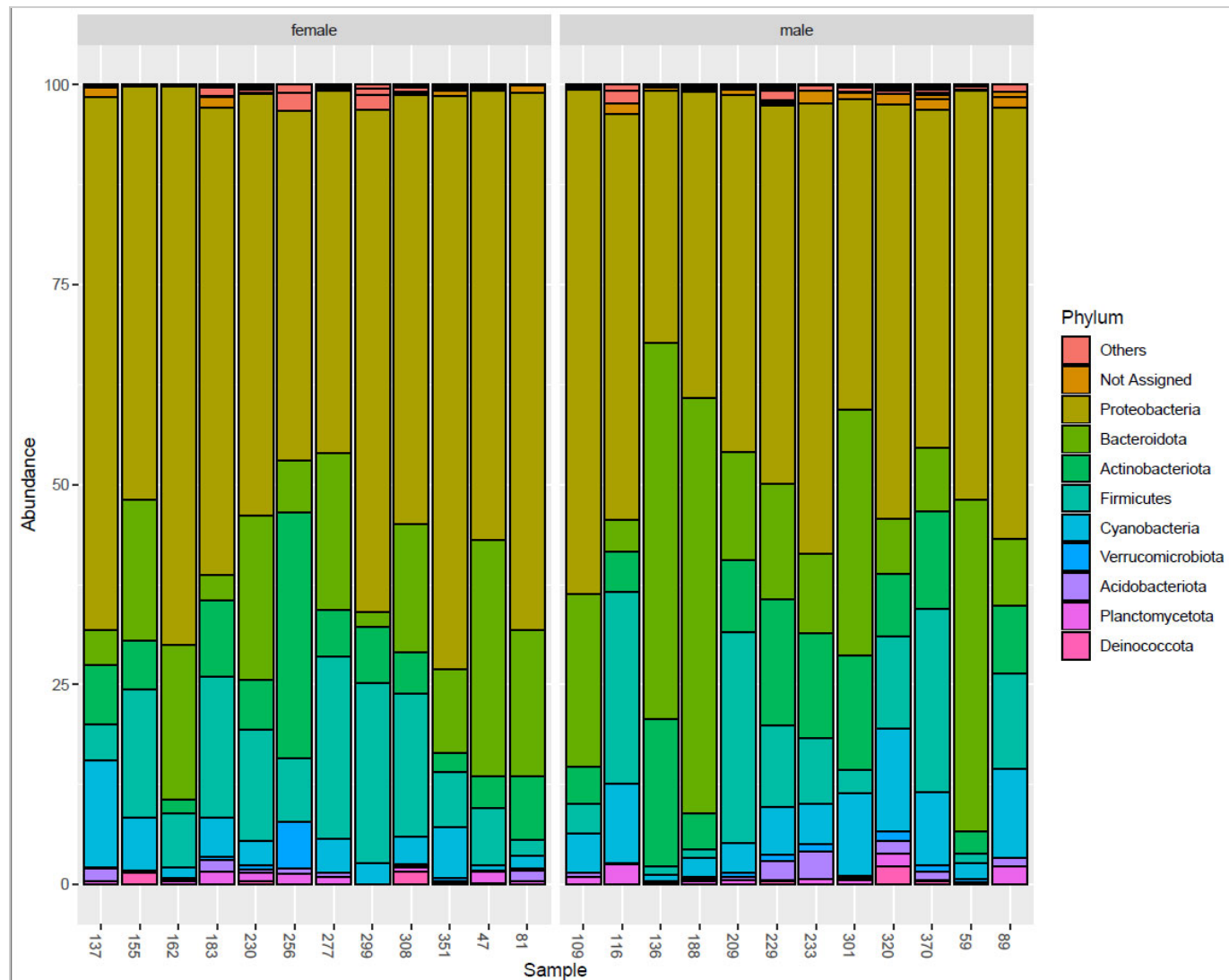
The phylum level is useful because it provides a broad overview of the community composition, while still capturing biological differences.

Why was your faceting choice useful?

Faceting by sex was useful because it allowed for direct visual comparison between male and female samples, making it easier to see any differences or similarities in microbial composition between them.

Upload screenshot: taxonomic bar plot

Choose File barchart_phylum_sex.png



Which taxa appear to be most abundant?

Looking at the taxa, it seems like Proteobacteria is the most abundant between the samples. It is followed by Bacteroidota.

Did taxonomic composition differ among groups?

There are some differences in relative abundances, but overall the taxonomic composition is fairly similar between the groups. Both males and females are dominated by the same phyla, that being Proteobacteria.

Part 12 — Overall interpretation

In 3–5 sentences, summarize what your DADA2 results, alpha diversity, beta diversity, and bar plot together suggest about this dataset.

DADA2 produced ASVs that seem to accurately represent the microbial communities present in the samples. Alpha diversity suggests that diversity varies more by sex than by population, with males generally showing higher richness and evenness. Beta diversity shows considerable overlap among groups, but some clustering indicates that sex has a moderate influence on community composition, while population differences are weaker. The taxonomic bar chart supports this by showing that communities are dominated by similar phyla

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset

Project/sample set: Section Report 3

Sequencing type: Illumina

Grouping variable: samples, population, sex, timepoint, bird, flock

Quality profiles

Purpose: The purpose of this step is to check the quality of the reads.

Reverse vs forward quality: Usually, the reverse-read quality is worse compared to forward-read quality.

Forward quality drop: Around 245bp

Reverse quality drop: Around 225bp

Forward truncation: 0 (Default since quality was still above 30)

Reverse truncation: 230

Why these values: I chose those truncation values (mainly just for the reverse) since that's where the quality dipped below 30.

filterAndTrim

Purpose: This step is important because it helps to trim the low-quality reads since they would impact the results in the future.

Parameters used:

Truncation at 230 for the reverse reads only. The front of those reads were good, it was just the end where it got a little iffy.

If trimmed too much: If you trim too aggressively, some tools may not work properly since there's not enough overlap (maybe speaking from experience)

If not trimmed enough: If you don't trim enough, the quality could be worse in the long run.

learnErrors

Purpose: LearnErrors makes it so DADA2 can recognize any sequencing errors, which can be seen on the plots it produces.

Model fit: Mostly yes

Why important: So learning errors rates is important before denoising because DADA2 relies on the error models to see if the differences are caused by actual biological variation or by sequencing errors.

dada + mergePairs

Denoising purpose: Denoising is correcting sequencing errors in the reads to identify and remove any substitutions, insertions, and any other mistakes introduced during sequencing.

ASVs vs OTUs: ASV interference is different from OTU clustering because ASV identifies sequence variants at single-nucleotide places whereas OTU clusters groups based on similarity.

mergePairs purpose: The forward and reverse reads are being merged to improve accuracy as they will have overlapping regions. This will allow errors to be detected and correctly to get a more reliable final sequence.

Need for overlap: The forward and reverse reads need to overlap to correct sequencing errors.

One reason reads fail to merge: If reads fail to merge, one likely reason is that there is not enough overlap between the reads.

Sequence table + sequence length distribution + chimera removal + taxonomy

Sequence table purpose: This table contains which organisms are present in the samples and their relative abundance.

Rows/columns/cells: The rows are individual samples. The columns are ASVs. The cell values represent the number of times an ASV was observed in the sample.

Sequence length distribution: The sequence length distributions shows a clean and short amplicon. It does have a bit more bars down the length, which suggests that there may be minor sequencing variation. It could also suggest chimeras.

Chimera removal purpose: A chimera is an artificial DNA sequence formed during PCR when fragments from different sequences are mistakenly joined together. This means that it should be removed because it doesn't actually represent a biological organism.

Why chimeras are a problem: Chimeras are a problem for interpretation because they mess with the results when they get picked up. Since they aren't actually any real biological organisms, they just create problems.

Taxonomy purpose: Taxonomy assignment adds biological meaning to the ASV table because it links each DNA sequence to known groups of organisms, like genus, species, phylum, etc.

Taxonomy limits: Some ASVs might not be assigned all the way to species because the reference database may not have an exact match.

Alpha diversity

Definition: Alpha diversity is a metric to represent to diverse a community is.

Observed: Observed measures the total number of species.

Shannon: Shannon measures species abundance between sets.

Simpson: Simpson measures works like Shannon, but is less sensitive to rare species.

Differences by population: Across all three metrics, migratory and resident populations show a fairly similar density overall. There is a lot of overlap between the two groups, but migratory samples have a sometimes higher Shannon and Simpson value. The difference doesn't seem to be too strong or consistent.

Differences by sex: Comparing alpha diversity by sex, there does seem to be a more noticeable difference. Males tend to have higher Observed richness, and slightly higher Shannon diversity compared to females. Simpson values are higher for both, but males appear to have more diversity and evenness.

Differences by flock: Looking at the flock, resident males show the highest diversity in Observed richness. Migratory males are also relatively high, but they seem to be slightly lower than resident males. Females (both migratory and resident) tend to have lower diversity overall, with more spread and some low outliers.

Beta diversity (NMDS Bray-Curtis)

Definition: Beta diversity is comparing the composition of one community to another.

Bray-Curtis: Bray-Curtis measures beta diversity, so if this metric is higher then there are no shared species. If it's lower, then there are more shared species.

NMDS: NMDS shows each community present and its distance from every other point.

Patterns by population: Migratory and resident samples show substantial overlap in ordination space, showing that their microbial communities are not strongly separated. While there is some loose clustering, the groups seem to be largely mixed.

Patterns by sex: Compared to population, there is moderate separation by sex. Male samples tend to cluster more tightly in one region (lower NMDS2 value, slightly higher NMDS1), while females are more dispersed and cover a wider range of the plot. This could potentially suggest that sex may influence microbial composition.

Patterns by flock: Looking at the flock, resident males seem to be the tightest cluster, indicating very similar

community composition. The migratory males are somewhat spread, but four points show partial grouping. Females, both migratory and resident, seem to be more dispersed across the plot.

Taxonomic bar plot

Taxonomic level: Phylum

Faceted by: sex

Why this taxonomic level? The phylum level is useful because it provides a broad overview of the community composition, while still capturing biological differences.

Why this faceting choice? Faceting by sex was useful because it allowed for direct visual comparison between male and female samples, making it easier to see any differences or similarities in microbial composition between them.

Most abundant taxa: Looking at the taxa, it seems like Proteobacteria is the most abundant between the samples. It is followed by Bacteroidota.

Did composition differ among groups? There are some differences in relative abundances, but overall the taxonomic composition is fairly similar between the groups. Both males and females are dominated by the same phyla, that being Proteobacteria.

Overall interpretation

DADA2 produced ASVs that seem to accurately represent the microbial communities present in the samples. Alpha diversity suggests that diversity varies more by sex than by population, with males generally showing higher richness and evenness. Beta diversity shows considerable overlap among groups, but some clustering indicates that sex has a moderate influence on community composition, while population differences are weaker. The taxonomic bar chart supports this by showing that communities are dominated by similar phyla across the groups (female and male), with only minor differences in relative abundance.